

P. Shubham Parashar

THEORETICAL PHYSICIST

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RESEARCH PROFILE

Theoretical condensed-matter physicist working on quantum transport, kinetic theory, electron hydrodynamics, magnetotransport, spatially resolved response, and quantum Hall systems. I develop controlled analytical and computational descriptions that connect microscopic kinetic structure to experimentally accessible observables and test where low-moment hydrodynamic closures cease to be complete.

SELECTED RESEARCH CONTRIBUTIONS

- **Kinetic transport beyond hydrodynamic closure.** Derived higher-gradient and finite-wave-number structure that exposes angular harmonics omitted by stress-level hydrodynamics, including a fourth-order closure obstruction and its chiral nonlocal completion.
- **Symmetry-selected kinetic observables.** Constructed a sixfold-driven local hydrodynamic null in which ordinary Ohmic and rank-two viscous dissipation vanish at the symmetry point while a higher kinetic harmonic survives as the leading local dissipative signal.
- **Finite-resolution transport inference.** Developed observable-level tests for thermoelectric Hall contrast and Hall-viscous response in scanning-probe measurements while accounting for boundary, higher-gradient, and finite-resolution nuisance channels.

EDUCATION AND RESEARCH

2022–2026	Ph.D., Physics , University of California San Diego. Advisor: Michael M. Fogler.
2017–2022	Researcher , University of California San Diego. Research with Benjamin Grinstein and Congjun Wu.
2016–2017	M.S., Physics , Indian Institute of Science, Bangalore. Advisor: Rahul Pandit.
2012–2016	B.S., Physics , Indian Institute of Science, Bangalore.

SELECTED PUBLICATIONS AND MANUSCRIPTS

Complete kinematic null for local kinetic dissipation in a sixfold-driven electron fluid. P. Shubham Parashar.
Under review at *Physica B* (2026).

Fourth-order closure obstruction and chiral nonlocality in circular kinetic magnetotransport. P. Shubham Parashar.
Under review at *Physical Review B* (2026). [arXiv:2607.22730](#).

Thermal and viscous contrast in quantum Hall scanning-probe images. P. Shubham Parashar.
Under review at *Physica E* (2026). [arXiv:2607.16315](#).

Diffusive and hydrodynamic magnetotransport around a density perturbation in a two-dimensional electron gas. P. Shubham Parashar and M. M. Fogler.
Physical Review B **113**, 155443 (2026). [arXiv:2601.16402](#).

Nanoscale imaging of magnetotransport around a circular p–n junction in graphene. Z. J. Krebs *et al.*, including P. Shubham Parashar.
Physical Review Letters **136**, 136301 (2026), Editors' Suggestion. [arXiv:2409.19468](#).

METHODS AND DISTINCTIONS

Methods Boltzmann kinetic theory · electron hydrodynamics · semiclassical transport · linear response · asymptotics · quantum transport · numerical modeling · Python / Mathematica / LaTeX

Distinctions Bhadra Fellowship (2024) · UCSD Scholar's Grant (2017) · KVPY Fellowship (2012–2017)

Updated 11 August 2026 Full publication record and research materials: [shubham-parashar-cmt.github.io](#)